

MANISHA CHAWLA

Lecturer - Computer Science, Artificial Intelligence & Data Science

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Date of Birth: 09/June/1993

PROFESSIONAL PROFILE

Computer Science educator and AI researcher with over 9 years of teaching, research and academic leadership experience. M.Tech (Computer Science) from IIT Bhilai with a perfect CGPA of 10/10 and recipient of the Senate Award. Research interests include Federated Learning, Split Learning, Distributed AI, Graph Neural Networks, Trustworthy AI, and Computer Vision, with publications in IEEE IoT Journal, MICCAI, and ICDM. Experienced in curriculum design, outcome-based education, research mentoring, industry collaborations, and interdisciplinary AI projects.

ACADEMIC QUALIFICATIONS

M.Tech - Computer Science & Engineering

Indian Institute of Technology (IIT) Bhilai, India

B.E. - Computer Science & Engineering

Government Engineering College, Bilaspur, India

TEACHING & ACADEMIC EXPERIENCE

Lecturer - Computer Science & Engineering

Government Girls Polytechnic, Bilaspur, India

Department of Skill Development & Technical Education, Government of Chhattisgarh

October 2016 - Present

- Taught core Computer Science and IT courses, including Operating Systems, Computer Networks, Data Structures, System Analysis and Design, C/C++/Python Programming, Cloud Computing, and IT Trends.
- Designed and delivered course plans, lab manuals, assessments, and project-based learning activities aligned with Outcome-Based Education (OBE).
- Mentored students in AI/ML-based projects, Python development and applied research.
- Contributed to NBA and AICTE accreditation, academic planning and quality assurance.

Academic & Administrative Responsibilities

- Academic In-Charge, Training & Placement In-Charge, Incubation Cell In-Charge, Research & Innovation Cell In-Charge, Examination, Laboratory, Website, Hostel, AICTE & NSS In-Charge roles

TEACHING ASSISTANT EXPERIENCE (IIT BHILAI)

- **CS559** - Computer System Design
- **CS550 / DSL501** - Machine Learning

- **DS250** - Data Analytics & Visualization
- **IC100** - Programming in C

RESEARCH PUBLICATIONS

Journal

- **Beyond Federated Learning for IoT: Efficient Split Learning with Caching and Model Customization**
IEEE Internet of Things Journal, Vol. 11, Issue 20, October 2024,
DOI: 10.1109/JIOT.2024.3424660

Conferences

- **Fair and Accurate Skin Disease Image Classification by Alignment with Clinical Labels**
MICCAI 2024, LNCS 15001 (Top ~11% acceptance rate)
DOI: https://doi.org/10.1007/978-3-031-72378-0_37
- **Entropy-Aware Training for Fast and Accurate Distributed Graph Neural Networks**
IEEE International Conference on Data Mining (ICDM 2023)
DOI: <https://doi.org/10.1109/ICDM58522.2023.00112>

Poster

- **AdaGradSelect: An adaptive gradient-guided layer selection method for efficient fine-tuning of SLMs**, *MFF CARE Conference on Data Science and Artificial Intelligence 2025*.

PATENTS

- **AI-Powered Predictive Analytics System for Early Detection of Cardiovascular Diseases**
Patent Publication, Intellectual Property Office, Government of India
Publication Date: **29 November 2024**
- **Vehicle Driver Drowsiness Detection Device**
Indian Design Patent, Design No. **437072-001**
Grant Date: **11 February 2025**
- **Cognitive Cloud Shield: Intelligent Anomaly Detection and Predictive Risk Management in Dynamic Cloud Environments**
Patent Publication, Intellectual Property Office, Government of India
Publication Date: **13 February 2026**

RESEARCH PROJECTS

- **Industry-Oriented Human Action Recognition for Smart Manufacturing and Industrial Safety**
(Collaborative project with Bokaro Steel Plant, Bhilai Steel Plant, and IIT Bhilai)
- **Distributed Graph Neural Networks for Scalable Large-Scale Data Analytics**
- **Adaptive Split Federated Learning for Robust and Communication-Efficient Distributed Training**
- **Bias Mitigation and Fairness Enhancement in AI Models for Healthcare Applications**
(Collaborative project with Shahid Hospital, Dalli Rajhara, and Jan Swasthya Kendra, Bilaspur)

REVIEWER EXPERIENCE

Professional Service: Reviewer for the IEEE Internet of Things Journal and MICCAI Conference

INDUSTRY & ACADEMIC CONSULTING EXPERIENCE

Independent Academic & AI Consultant (Freelance)

Remote | Ongoing

- Consulting in Artificial Intelligence, Machine Learning, and Data Science.
- Research prototyping, model development, and experimentation.
- Academic writing, technical documentation, and project mentoring.
- Collaboration on industry- and research-oriented AI solutions.

FACULTY DEVELOPMENT & PROFESSIONAL TRAINING

- **Contributing Faculty**, AICTE Faculty Development Programs hosted by NIT Hamirpur (January 2025) and VIT Vellore (September 2023) on advanced themes in IoT, Intelligent Systems, and Artificial Intelligence; conducted a hands-on expert lecture on Federated Learning with live demonstrations.
- **Participant**, AICTE-ATAL Faculty Development Programs and workshops covering:
 - Artificial Intelligence, Machine Learning, and Data Science
 - Deep Learning, Computer Vision, and AI for Healthcare
 - Industry 4.0, Industrial IoT, and Smart Mobility
 - Cyber Security, Quantum Computing, and 5G/6G Technologies
 - Entrepreneurship Development and emerging technologies

AWARDS & HONORS

- Bank of Baroda Achiever Award
- Nominee - Director's Gold Medal, IIT Bhilai
- Senate Award, IIT Bhilai
- University Topper
- State Rank Holder - Senior Secondary (Mathematics Group)
- GATE Qualified - 5+ times

ACADEMIC LEADERSHIP & REPRESENTATION

- Presenter, 19th Chhattisgarh Young Scientists Congress (CYSC-2024)
- Represented IIT Bhilai at India Mobile Congress (IMC 2023)
- Sci-Tech Event Coordinator, MERAZ 2022 (IIT Bhilai)
- Participant, Inter-IIT Tech Meet 12.0 (Adobe Challenge)
- Participant, Dark Patterns Buster Hackathon 2023
- Shortlisted Team Member, Smart India Hackathon 2023

PROFESSIONAL SERVICE & COMMUNITY ENGAGEMENT

- NSS Program Officer, Government Girls Polytechnic Bilaspur, India.
- Active involvement in student outreach, environmental awareness, and community initiatives.